# TD 9: Clausal Logic and the Resolution Method

## Exercise 1

1. Convert these formulas into Clause Set notation (e.g., $\{\{L_1, L_2\}, \{L_3\}\}$):
   - $(P \vee Q) \wedge (\neg R \vee S) \wedge P$
   - $(\neg A \vee B \vee C) \wedge (A \vee \neg C)$
   - $P \wedge \neg P$
   - $\neg((P \wedge Q) \Rightarrow R)$
   - $P \Leftrightarrow Q$
   - $\neg(\neg P \wedge (Q \vee R))$
   - $(P \Rightarrow Q) \wedge \neg R$
2. Determine whether the set $\Gamma = \{P \vee Q, \neg P, \neg Q\}$ is satisfiable.
3. Is the clause set $\{P, \neg P \vee Q, \neg Q\}$ consistent? Justify your answer.

## Exercise 2

Consider the following logical argument: "If it rains ($R$), then the ground is wet ($W$). It is currently raining. Therefore, the ground is wet."
1. Represent the premises $\Gamma$ and the conclusion $\psi$ in propositional logic.
2. To prove $\Gamma \vDash \psi$ using the refutation method, we must show that a specific set of formulas $\Phi$ is unsatisfiable. Define $\Phi$ in terms of $\Gamma$ and $\psi$.
3. Convert the set $\Phi$ into a set of clauses (Conjunctive Normal Form). 4. Using the resolution rule: $\frac{C_1 \cup \{L\}, \quad C_2 \cup \{\neg L\}}{C_1 \cup C_2}$, show the derivation that leads to the empty clause ($\square$).
5. In the context of the Davis-Putnam or Resolution procedure, what is the semantic meaning of deriving the empty clause ($\square$), and why does it terminate the proof?

## Exercise 3

Formalize the following English argument and convert it into a single Clause Set for a refutation proof: "If the file is open ($O$), the lock is active ($L$). The file is open. Prove: The lock is active."

## Exercise 4:

A. Which of the following are valid?

1. If $\Gamma, \alpha \vDash \sigma$, then $\Gamma, \alpha \vee \beta \vDash \sigma$.
2. If $\Gamma, \alpha \vDash \sigma$, then $\Gamma, \alpha \wedge \beta \vDash \sigma$.
3. $\alpha \leftrightarrow \beta \vDash \alpha \wedge \beta$

B. Assuming that $\Gamma$ is satisfiable, which of the following propositions are valid?

1. If $\Gamma \nvDash \beta$, then $\Gamma \cup \{\beta\}$ is not satisfiable.

2. If $\Gamma \cup \{\beta\}$ is not satisfiable, then $\Gamma \nvDash \beta$.
3. If $\Gamma \vDash \beta$, then $\Gamma \cup \{\neg\beta\}$ is satisfiable.
4. If $\Gamma \vDash \beta$, then $\Gamma \cup \{\neg\beta\}$ is unsatisfiable.
5. If $\Gamma \cup \{\neg\beta\}$ is not satisfiable, then $\Gamma \vDash \beta$.
6. If $\Gamma \cup \{\neg\beta\}$ is not satisfiable, then $\Gamma \nvDash \beta$.

## Exercise 5

A. A resolvent is a new clause derived from two "“parent” clauses that contain a complementary pair of literals (e.g., $P$ and $\neg P$). Find all possible resolvents for the following pairs:

1. $C_1 = \{P, Q, R\}, \quad C_2 = \{\neg Q, S\}$
2. $C_1 = \{P, \neg Q\}, \quad C_2 = \{Q\}$
3. $C_1 = \{P, Q\}, \quad C_2 = \{\neg P, R\}$
4. $C_1 = \{\neg P, \neg R\}, \quad C_2 = \{P, R\}$ *(Note: Can you resolve both at once?)*

B. Using the resolution rule, derive the final conclusion for the following sets. If you reach the empty clause ($\square$), state what this implies about the set.

1. $\Gamma = \{P, \quad \neg P \vee Q, \quad \neg Q \vee R\}$ (Derive the new clause)
2. $\Sigma = \{P \vee Q, \quad \neg P, \quad \neg Q\}$ (Resolve until the end)

## Exercise 6

To prove an argument is valid ($Premises \models Conclusion$), we must negate the conclusion and show the resulting Clause Set leads to $\square$. Prove:

1. Syllogism: $\{P \Rightarrow Q, \quad Q \Rightarrow R\} \models P \Rightarrow R$
2. Case Analysis: $\{(P \vee Q), \quad (P \Rightarrow R), \quad (Q \Rightarrow R)\} \models R$
3. Simple Tautology: Show $(P \wedge Q) \Rightarrow Q$ is valid using resolution.

## Exercise 7

Try to use resolution to prove: $\{P \Rightarrow Q, \quad Q\} \models P$.

- Observe what happens during the resolution process. Can you derive $\square$?
- What does this result tell you about the “Fallacy of Affirming the Consequent”?

## Exercise 8

1. Draw a tree for $S = \{\{P, Q\}, \{\neg P, Q\}, \{P, \neg Q\}, \{\neg P, \neg Q\}\}$ showing how it reduces to $\square$.

2. Why can different resolution "orders" (which clauses you pick first) lead to different proof speeds?

3. What does it mean for the original formula if you have exhausted all possible resolutions but never derived $\square$?

## Exercise 9

A. Given the clause set: $\Gamma = \{\{p, q\}, \{\neg p, r\}, \{\neg q, r\}, \{\neg r, s\}, \{\neg s\}\}$. Use resolution to determine whether $\Gamma$ is satisfiable.
B. Prove using resolution that: $\{(p \to q), (q \to r), p\} \models r$
C. Check whether: $\{p \vee q,\ \neg p \vee r\} \models q \vee r$ using resolution.
D. Consider: $\Gamma = \{\{p, q\}, \{\neg p, q\}, \{p, \neg q\}, \{\neg p, \neg q\}\}$

1. Show that $\Gamma$ is unsatisfiable.

2. Find a minimal unsatisfiable subset.

E. Let: $\Gamma = \{p \to q,\ q \to r,\ r \to s,\ p\}$

1. Use Theorem 2 to show $\Gamma \models s$ (without resolution).

2. Then prove the same using resolution.

Theorem 2: $\Gamma \models \alpha$ and $\Gamma \models (\alpha \to \beta) \implies \Gamma \models \beta$

## Exercise 10

Demonstrate the following theorems:
**Theorem 1**: $\alpha_1, \alpha_2, \ldots, \alpha_n \models \beta$ if and only if $\alpha_1, \alpha_2, \ldots, \alpha_{n-1} \models \alpha_n \to \beta$
**Theorem 2**: Let $\Gamma$ is a set of formulas. If $\Gamma \models \alpha$ and $\Gamma \models \alpha \to \beta$ then $\Gamma \models \beta$
**Theorem 3**: If $\Gamma \models \beta$ and $\Gamma \models \neg\beta$ then $\Gamma$ is unsatisfiable.
**Theorem 4**: $\alpha_1, \alpha_2, \ldots, \alpha_n \models \beta$ if and only if $\{\alpha_1, \alpha_2, \ldots, \alpha_n, \neg\beta\}$ is unsatisfiable.

## Exercise 11

Demonstrate the decidability theorems: Because the number of possible clauses over a finite set of variables is finite, the resolution procedure always terminates in propositional logic.